Vol 4 Chapter 7 — Inflation Parameters

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2026-05-23 Saturday

Chapter 7 — Inflation Parameters

Cosmic inflation — the period of rapid exponential expansion in the early universe — leaves observable signatures in the CMB. The tensor-to-scalar ratio r, the tensor spectral index n_t, and the running α_s of the scalar spectral index are the canonical inflation observables.

7.1 Tensor-to-scalar ratio r

BST predicts $r \approx 0$ at observable precision — the substrate's inflation scale $T_c \ll m_{\rm Pl}$ suppresses primordial gravitational waves below current detection thresholds. Current experimental bound: $r < 0.06$ (Planck + BICEP/Keck). BST is consistent.

A future detection of $r > 0.01$ would refute the substrate prediction and require revision.

7.2 Scalar-spectral-index running

$$\alpha_s = -(n_s - 1)^2 = -\left(\frac{n_C}{N_{\max}}\right)^2 = -\frac{25}{18769} \approx -1.33 \times 10^{-3}.$$

Planck measurement: $\alpha_s = -0.0045 \pm 0.0067$ — substrate prediction within 0.5σ.

7.3 Inflation scale

The substrate's natural inflation scale is set by Casimir + Mersenne integer combinations, with $T_c \sim 10^{15}$ GeV consistent with single-field slow-roll inflation models.

7.4 What comes next

Chapter 8 — BBN element abundances — derives the primordial light-element abundances and the substrate resolution of the Li-7 problem.

Where to look this up: Inflation parameter battery: Task #85. For standard inflation: Liddle and Lyth, *Cosmological Inflation and Large-Scale Structure*.